

Securely Connecting Amazon Quick to Internal Systems

Private MCP Connection — bringing AI inside the enterprise security perimeter

The AI assistant uses sensitive data, yet the data never leaves the private network.

Agenda

1. **Context** — Amazon Quick and the MCP connector
2. **The Problem** — why we can't expose internal data
3. **The Solution** — Private MCP Connection
4. **How It Works** — the request flow
5. **Use Case** — incident investigation
6. **Business Value** — what we gain
7. **Layered Security** — network + identity
8. **Why This Approach** — vs. alternatives
9. **Summary & Next Steps**

Context

Amazon Quick is an enterprise AI assistant: search data, build agents, automate workflows.

Quick connects to external systems through the **MCP connector**.

But many critical systems live **inside a private network**:

- Internal databases
- Monitoring / logging systems
- Custom operational tools

How can AI use this data without exposing it to the Internet?

The Problem

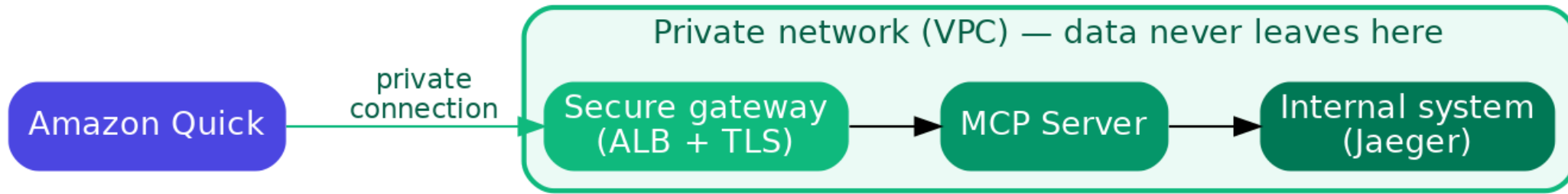


Problem: sensitive data travels across the public Internet

Opening a public endpoint for the AI to call introduces:

- Sensitive data traveling over public paths
- A larger attack surface
- Difficulty meeting data-security compliance

The Solution: Private MCP Connection



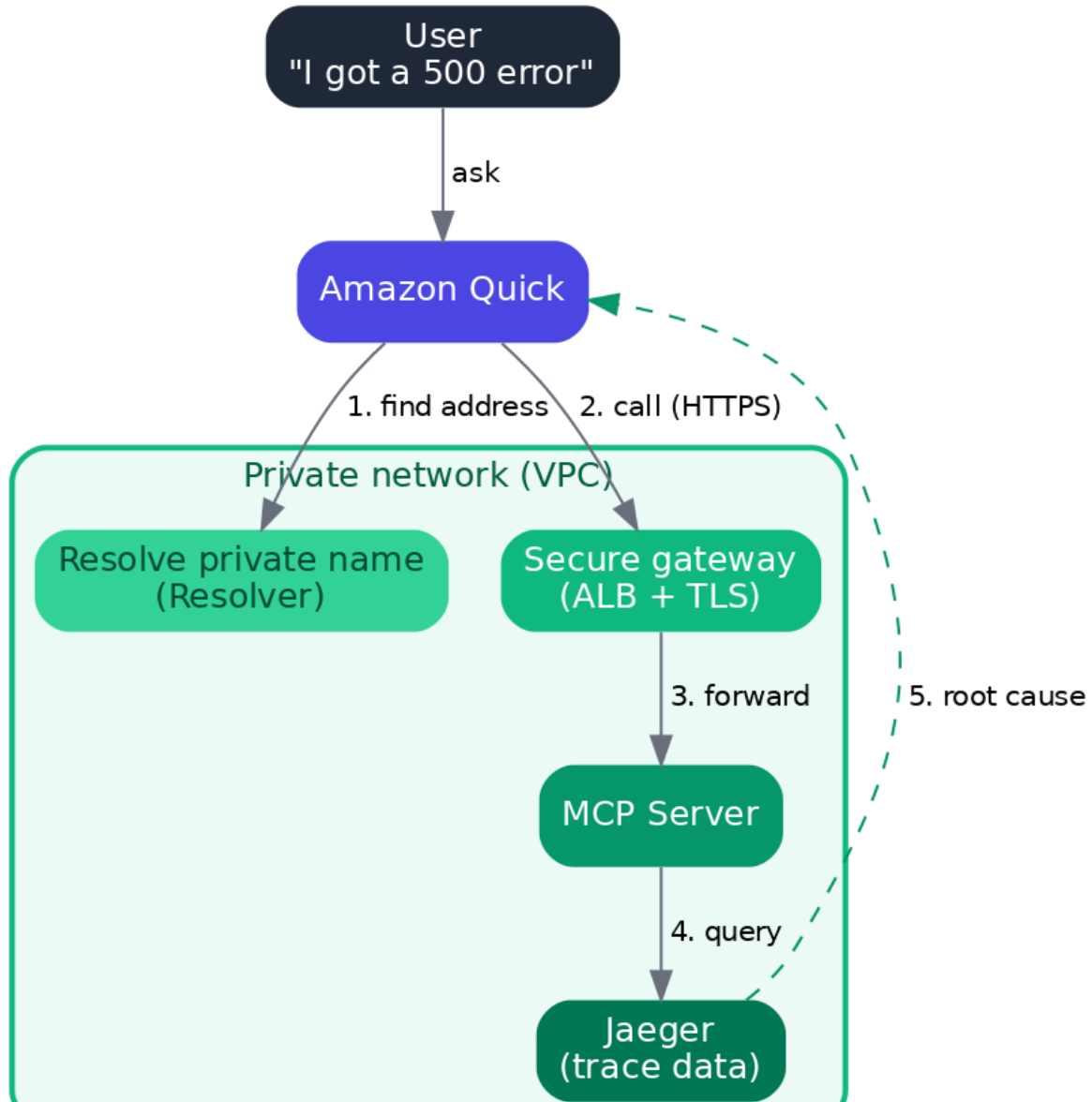
Solution: Private MCP — AI reaches data inside the security perimeter

Two core principles:

- **Public certificate** → connection is properly encrypted (valid TLS)
- **Private hostname** → only visible inside the internal network

→ AI can use the data; the data **never leaves**.

How It Works



Use Case: Incident Investigation

Scenario: A customer reports *"I got a 500 error at checkout."*

Before	With this solution
Engineers dig through many logs	Ask Quick in plain language
Time-consuming	Quick calls MCP inside the private net
Easy to miss things	Root cause found automatically

Result: *"The payment service timed out calling the bank API"* — in seconds.

Business Value

Aspect	Benefit
Security	Data never leaves the network; no public endpoint
Compliance	Meets sensitive-data isolation requirements
Speed	Natural-language incident response, instantly
Simplicity	Standard networking the team already knows
Cost	Fewer moving parts, no extra data-processing fees

Layered Security

Layer 1 — NETWORK protection (always on)

Only in-VPC components can reach it

Hostname resolves privately only

Tight per-component permissions

Why This Approach

Compared with going through a managed gateway service:

- **Fewer components** → easier to operate, fewer failure points
- **No public ingress** → better security by default
- **Amazon Quick's first-class path** → stable and supported
- **Flexible authentication** → run with no auth (network isolation) or enable OAuth

Summary

More useful AI — it works with real data

Safer data — it never reaches the Internet

Simpler operations — standard networking

AI meets data, securely.